

A Coupling-Class Impossibility Argument for Intact Macroscopic Condensed Matter at $\beta \approx 0.99$ in Ordinary Interstellar Environments

— *or, Why Rocks Do Not Go 0.99c*

Allen Hall

Draft 2, prepared 2026-07-09, revised after a second adversarial review round (corrections logged in Appendix B). Verification script: `calcs.py` (this repository). All numerical claims in this paper are reproduced by that script; line-by-line chains appear in Appendix A.

Abstract

We argue that, under known physics, no demonstrated or physically credible pathway — natural or engineered — accelerates an ordinary macroscopic condensed body (an intact solid ≥ 1 cm of ordinary composition) to $\beta = 0.99$ ($\gamma \approx 7.09$) and transports it intact through ordinary interstellar matter over astrophysical distances. The claim is deliberately conditional and its modality is tiered: every load-bearing step is tagged as a physical contradiction [I], an astrophysical rarity bound [P], or an engineering-scale closure [E], and the conclusion inherits exactly the strength of its weakest necessary tag — this is a coupling-class impossibility argument under stated environmental assumptions, not an unconditional theorem. The argument is a proof by cases over an exhaustive classification of physical couplings. Gravity, the unique coupling that accelerates a neutral body without stress-bearing internal force paths, lacks any known or physically credible ejection channel to $\gamma \approx 7$ for strength-bound solids: the one channel that ejects (bound-orbit exchange) is capped near the ISCO orbital speed $\sim 0.5c$ per encounter, and chains die dynamically — after one boost the body arrives at every subsequent system too fast for the exchange mechanism to grip it. Every non-gravitational coupling delivers momentum through material response, and each faces a fatal conjunction among energy scale (5.5×10^{17} J/kg), material dissipation, beam diffraction, thermal rejection, shielding mass, and relativistic environmental flux. For beamed sails we show the $14\times$ Doppler factor of the burn is a conserved burden: a fixed-frequency system imposes it on the sail's absorptivity, a chirped transmitter relocates it onto vacuum-ultraviolet source and optics physics, and no measured material class carries it at either end. At $0.99c$ the interstellar medium at ordinary densities ($n \geq 0.1$ cm $^{-3}$) imposes ~ 0.2 – 2 MW/m 2 of frontal heating — equilibrium temperatures at or above the failure point of every ordinary solid — while cascade-stopping shielding costs $\sim 10^6$ times any beam-driven mass budget, and active (magnetic) shielding fails on the neutral gas and dust that dominate the damage. Trajectories confined to rarefied phases (hot bubbles, halo, intergalactic space) escape the thermal kill and are owned as boundary cases, as are eon-timescale burns and artificial vacuum corridors: each lands on an [E]- or [P]-level closure, stated as such. We survey prior art: components of this argument exist in the literature — relativistic rocket energetics (Semyonov 2014, 2018), interstellar-medium damage to relativistic spacecraft (Hoang et al. 2017; Drobny et al. 2021), laser-sail scaling ceilings conceded by the pathway's own advocates (Lubin 2016), gravitational ejection ceilings for stars (Guillochon & Loeb 2015), and a formal environmental speed limit binding only at $\gamma \sim 10^8$ (Yurtsever & Wilkinson 2018). In a targeted, adversarially verified search we found no prior source that assembles these constraints into a coupling-class impossibility argument for ordinary macroscopic condensed bodies at $\beta \approx 0.99$; each source we found stops short of impossibility, addresses a different speed regime, or addresses non-macroscopic bodies. Nature's own record corroborates the argument: every observed mechanism that accelerates matter to relativistic speeds delivers it as particles or plasma; the fastest observed intact bodies are stars below a percent of c , the theoretical stellar ejection ceiling is $c/3$, and the fastest known solid objects

move at $10^{-4} c$.

1. Statement of the problem

1.1 The claim

Claim (central). *Under known physics, no demonstrated or physically credible pathway — natural or engineered — both accelerates an ordinary macroscopic condensed body to $\beta = 0.99$ and transports it intact through ordinary interstellar matter over astrophysical distances. Gravity avoids heating but has no known or credible ejection channel to $\gamma \approx 7$ for strength-bound solids. Every non-gravitational pathway delivers momentum through material couplings — radiation pressure, reaction mass, particle flux, or contact/field structures — and each faces at least one fatal conjunction among energy scale, material dissipation, beam diffraction, thermal rejection, shielding mass, and relativistic environmental flux. This is not a proof against arbitrarily large artificial vacuum accelerators or against new physics; those cases are bounded explicitly in §9 and carry [E]-level closures only.*

An earlier draft stated this as an unconditional "no-go theorem" over the bare act of acceleration. That was too strong, in two ways this draft repairs: the interstellar-medium tier (§5) binds *transport through ordinary environments*, not acceleration in an engineered vacuum; and several closures are engineering-scale [E] or rarity [P] statements, which cannot support the same word as a physical contradiction [I]. The claim above, and every section below, now says which it is.

Definitions. Five terms carry weight.

Ordinary macroscopic condensed matter. A solid body of characteristic linear size $L \geq 10^{-2} \text{ m}$, electrically neutral, bound by chemical (electromagnetic) cohesion, with tensile/compressive strength $S \leq 10^{10} \text{ Pa}$ (diamond is $\sim 10^{11} \text{ Pa}$ in ideal crystals; engineering solids are 10^7 – 10^9 Pa), density $\rho \sim 10^3$ – $2 \times 10^4 \text{ kg/m}^3$, and melting or sublimation temperature $T_m \leq 4,300 \text{ K}$ (the highest known, Ta–Hf–C systems). "Intact" means the body arrives as a connected solid, not as vapor, plasma, or a debris cloud.

Payload, propulsion structure, accelerated artifact. The *payload* is the ordinary condensed body required to arrive intact. The *propulsion structure* is everything attached to move it — sail, shield, radiator, propellant, field coils — and may be expendable. The *accelerated artifact* is payload plus attached structure during the burn. The claim binds the **payload**; but every constraint on the artifact during acceleration (areal mass, temperature, stress) binds the payload through whatever must be attached to it at that moment, and we state at each step which mass is being constrained.

$\beta = 0.99$. Then $\gamma = (1 - \beta^2)^{-1/2} = 7.0888$, and the kinetic energy per unit rest mass is

$$E_k/m = (\gamma - 1)c^2 = 6.089 c^2 = 5.47 \times 10^{17} \text{ J/kg}.$$

Every kilogram must receive 6.09 times its rest-mass energy as kinetic energy. This single number is the antagonist of every argument below.

Ordinary interstellar matter. Gas and dust at the densities that fill the great majority of galactic-disk sightlines: $n \sim 0.1$ – 100 cm^{-3} with $\sim 1\%$ of mass in grains. Rarefied phases (hot ionized bubbles, halo, intergalactic space, $n \leq 10^{-2} \text{ cm}^{-3}$) exist and are treated as the boundary cases they are (§5.1, §9.5), not ignored.

Known physics. The Standard Model plus general relativity, together with the measured properties of condensed matter (strengths, absorptivities, vapor pressures) and the measured astrophysical environment

(interstellar densities, black-hole populations and velocities). We claim no prophecy about undiscovered physics.

1.2 The form of the argument, and what each word is allowed to mean

The claim is not derived from a single conservation law. It is a **scaling-emergent impossibility argument**: each requirement in isolation can be met; the conjunction cannot. The proof form is a case analysis over an exhaustive classification of couplings (§3), where each case is closed by an exhibited contradiction, an explicit rarity bound, or an exhibited artifact scale — never by rhetoric.

Epistemic ground rules. Every load-bearing claim is tagged, and each tag licenses specific language and no more:

Tag	Meaning	Licensed language
[I]	physics-derived contradiction (two inequalities jointly unsatisfiable)	"forbidden," "impossible"
[P]	astrophysical rarity bound	"rate-suppressed," "effectively absent," never "impossible"
[E]	no principle violated; required artifact scale exhibited and quantified	"not credible for bounded engineering actors," never "forbidden"

The central claim inherits the weakest necessary tag of each branch: gravitational pathways close at [I]-per-channel + [P]-on-chains; transport through ordinary ISM closes at [I]; engineered acceleration in artificial environments closes at [E]. A reader who wants only theorems should read §5 and §8.1–8.2 and stop; a reader who wants the full case structure gets the tags.

Observational absence is assigned zero evidentiary weight throughout (§11 explains why); the argument rests on dynamics, not on surveys.

2. The energy scale and the two fates of transferred energy

2.1 Kinematic anchors

β	γ	E_k/mc^2	$E_k \text{ (J/kg)}$
0.10	1.0050	0.0050	4.53×10^{14}
0.20	1.0206	0.0206	1.85×10^{15}
0.50	1.1547	0.1547	1.39×10^{16}
0.90	2.2942	1.2942	1.16×10^{17}
0.99	7.0888	6.0888	5.47×10^{17}

Between 0.5c and 0.99c the required energy grows by a factor of 39. The argument's structure follows this cliff: below ~0.5c one heat-free pathway exists in principle (§4); above it, none does.

2.2 A correction we impose on ourselves: transferred energy is not deposited heat

A tempting shortcut — "any process transferring 6 mc^2 to a body must thermalize a ruinous fraction of it in the body" — is **false**, and the argument must not lean on it. The counterexample is the beamed light sail: a photon of lab energy E reflecting from a mirror receding at β returns with energy $E(1-\beta)/(1+\beta)$; the fraction transferred

to the sail's kinetic energy is $2\beta/(1+\beta) \rightarrow 0.995$ at $\beta = 0.99$. Reflected energy leaves at c and never thermalizes in the sail. Only the *absorbed* fraction α heats it, and demonstrated dielectric mirrors reach $\alpha \sim 10^{-6}$ at their design wavelength (ion-beam-sputtered stacks; cavity finesse $> 10^5$ is routine). Integrating the reflection efficiency over the burn, the incident beam energy needed to reach $0.99c$ is

$$E_{\text{inc}} = \int_0^{0.99} (1+\beta)/(2\beta) c^2 d\gamma = 5.89 \times 10^{17} \text{ J/kg} \approx 1.08 \times E_k,$$

so at $\alpha = 10^{-6}$ the absorbed dose is only $\sim 5.9 \times 10^{11} \text{ J/kg}$. A sail of areal density 1 g/m^2 ($1,000 \text{ m}^2$ of sail area per kilogram, radiating from two faces) at $1,000 \text{ K}$ rejects heat at $2\sigma T^4 \times 1000 \text{ m}^2/\text{kg} = 1.13 \times 10^8 \text{ W/kg}$ and sheds that dose in $\sim 5.2 \times 10^3 \text{ s}$. **The naive total-energy thermal budget closes at the design wavelength.** [I — as a counterexample]

This is why the argument must be, and is, built from the sharper facts: the burn's $14\times$ Doppler burden must land somewhere in the photon chain and no material class carries it (§6), the beam cannot stay focused for the required time (§7), ordinary interstellar matter is itself a beam (§5), and the sole coupling exempt from all of this has no credible ejection channel (§4). The remainder of the paper assembles exactly these.

3. Classification of couplings (the case structure)

Lemma 1 (exhaustiveness). *Any process that changes the momentum of a body couples to it through one of:*

1. **(G)** gravity — spacetime curvature acting on mass-energy;
2. **(EM-γ)** electromagnetic radiation — photons absorbed, emitted, or reflected at the body's surface;
3. **(EM-s)** structured electromagnetic force — static or quasi-static fields acting on charges/currents/magnetization carried by the body (coilguns, mass drivers, charged-body accelerators);
4. **(M)** matter exchange — momentum carried by massive particles: rocket exhaust ejected from the body, or external particle beams / medium impinging on it;
5. **(C)** contact — mechanical push through a structure (a limiting case of EM-s at atomic scale; treated with it).

Justification. The Standard Model admits four interactions. The strong and weak forces have ranges of 10^{-15} – 10^{-18} m and cannot exert coherent macroscopic forces on a neutral body; they enter only inside case (M) as the mechanism of particle energy deposition. What remains is gravity and electromagnetism in its radiative, quasi-static, and particle-mediated guises. The list is exhaustive under known physics. ■

Lemma 2 (the structural dichotomy). *For ordinary electrically neutral condensed bodies, gravity is the only coupling that accelerates all mass-energy without stress-bearing internal force paths. A body in free fall is locally inertial at any coordinate acceleration whatsoever — 1 g or 10^{12} g — by the equivalence principle: no load path, no internal stress, no compression, no heat. Every other coupling acts on the body's charges: the force enters at surfaces or through the bulk, is redistributed by elastic stress, and is subject to the dissipative response of real materials (finite conductivity, finite absorptivity, hysteresis). The exemption and the equivalence principle are the same fact. Only tidal *gradients* — the part of gravity that cannot be transformed away — stress the body, and they are bounded in §4.*

Narrowing, stated explicitly: electromagnetic acceleration can approximate a volumetric body force — but only by making the object a charged or current-bearing engineered system (uniform charging, embedded currents, superconducting or magnetized structure), at which point Coulomb self-stress, field-emission limits on surface charge ($E \lesssim 10^9 \text{ V/m}$ caps charge-to-mass near 10^{-5} C/kg for meter-scale solids, giving accelerations no better than the §8.3 strength cap), ohmic loss, quench, and plasma formation enter, and the object is no longer an

ordinary neutral solid. Such systems are closed inside cases (EM-s) and (M); the loophole is named so it cannot be reopened silently. ■

The case analysis is then: (G) in §4; (M)-**environmental** — the interstellar medium, which every pathway must face — in §5; (EM- γ) in §6–7; (M)-**rocket** and (EM-s/C) in §8.

4. Case G: gravity has no credible ejection channel to $\gamma = 7$ [I per channel; P on chains]

Gravity is the one loophole candidate: it accelerates without heating. The question is dynamical: *by what mechanism could gravitational dynamics eject an intact strength-bound body, and how fast?* An earlier draft claimed a universal ISCO-based kinematic cap at $\sim 0.5c$; adversarial review showed that claim proves less than it asserted (chained boosts compound in rapidity, not velocity — §4.3 owns this), and this section now closes the case channel by channel. The conclusion survives; the warrant is repaired.

4.1 The channels, enumerated

Gravitational interactions with a body divide cleanly:

1. **Single flyby of any black hole (including hyperbolic passes inside the ISCO radius).** Elastic in the hole's rest frame: $v_{\infty \text{ out}} = v_{\infty \text{ in}}$. A hyperbolic trajectory *may* pass deep inside r_{ISCO} (down to the capture threshold) and reach locally enormous speeds at pericenter — and leaves with exactly the speed it came with. Local pericenter speed is not terminal speed at infinity. Gain requires the hole to be *moving*: channel 2. [I — energy conservation]
2. **Flyby of a moving black hole ("gravitational mirror").** Lab-frame gain per pass is bounded by $\sim 2u$, twice the hole's own velocity. The measured black-hole velocity distribution caps u : stellar-mass holes move at galactic speeds ($\leq 10^{-3} c$); the fastest known compact-object kicks — gravitational-wave recoil of merger remnants — reach $\sim 5,000 \text{ km/s} \approx 0.017c$. One pass gains $\leq 0.03c$; reaching $0.99c$ this way requires a relay of hundreds of maximally kicked, freshly merged holes arranged along a trajectory. [I on per-pass gain from the measured velocity distribution; P on the relay]
3. **Ejection from a bound state (Hills-type exchange, binary disruption).** The one channel that manufactures large ejection speeds from rest. Terminal speed is of the order of the orbital speed at the deepest survivable *bound* orbit — the body must be temporarily bound to be exchanged — and stable bound orbits end at the ISCO. This is the channel §4.2 bounds.
4. **Penrose-type energy extraction.** Closed in §4.2 (Spin).

Channel 3's logic is established in the literature for stars. Hills (1988) found stellar-binary disruption by a supermassive black hole ejects stars at up to $\sim 4,000 \text{ km/s}$; Guillochon & Loeb (2015), asking precisely the maximum-ejection question ("the fastest unbound stars in the universe"), find the classical mechanism tops out near 10^4 km/s ($\sim 0.03c$) and that only the extreme configuration — a star bound to the secondary of a merging massive-black-hole binary — occasionally reaches $\sim 10^5 \text{ km/s}$, **one third the speed of light**, the ceiling being set by tidal disruption of the star at its deepest survivable pericenter. (The observed record is S5-HVS1 at $1,755 \text{ km/s} \approx 0.006c$; Koposov et al. 2020.) What follows transposes their tidal-limit logic from self-gravity-bound stars to strength-bound solids, which survive far deeper — and finds the per-encounter scale that replaces $c/3$.

4.2 Channel 3's per-encounter scale: ISCO, and where strength bites

Two constraints set the floor:

(a) **Dynamics.** Inside the innermost stable circular orbit (ISCO), $r_{\text{ISCO}} = 6GM/c^2$ for a Schwarzschild hole,

there are no stable orbits: a body that dips inside without precisely tuned angular momentum plunges. The locally measured circular-orbit speed at the Schwarzschild ISCO is

$$v_{\text{ISCO}} = c/2, \quad \text{independent of the black-hole mass.}$$

(b) Material strength. The tidal gradient $2GM/r^3$ across a body of half-length L loads it with stress $\sigma_t \approx \rho (2GM/r^3) L^2/2$. Survival requires $r^3 \geq \rho G M L^2/S$. For a 1-m rock ($\rho = 3,000 \text{ kg/m}^3$, $S = 10^8 \text{ Pa}$):

M_BH	r_strength	r_ISCO	governing	local v_circ at limit
10 M $\odot$	215 km	88.6 km	strength	0.28 c
10 ³ M $\odot$	999 km	8.9 $\times 10^3$ km	ISCO	0.50 c
10 ⁶ M $\odot$	10 ⁴ km	8.9 $\times 10^6$ km	ISCO	0.50 c
4.3 $\times 10^6$ M $\odot$ (Sgr A*)	1.6 $\times 10^4$ km	3.8 $\times 10^7$ km	ISCO	0.50 c

Correction to an earlier draft: at the ISCO of a *stellar-mass* hole the tidal gradient is $\sim 4 \times 10^6 \text{ s}^{-2}$ per meter — a meter-scale rock is shredded there ($\sigma_t \sim 1.4 \times 10^9 \text{ Pa} \geq S$); an earlier draft's figure of "a few hundred Pa" was wrong by seven orders of magnitude. The conclusion nevertheless *strengthens*: because $r_{\text{strength}} \propto (ML^2)^{1/3}$ while $r_{\text{ISCO}} \propto M$, the tidal gradient at the ISCO falls as $1/M^2$, and a meter rock reaches the ISCO with margin at every $M \geq 10^3 \text{ M}\odot$ (tidal stress $\sim 10^5 \text{ Pa}$ at $10^3 \text{ M}\odot$, $\sim 0.1 \text{ Pa}$ at $10^6 \text{ M}\odot$ — margins of 10^3 to 10^9 against $S = 10^8 \text{ Pa}$). Tidal stress scales as L^2 : it shreds stars (self-gravity-bound, effective $S \rightarrow 0$ at scale) and ignores rocks. The deepest survivable *bound* configuration in the universe — a small strength-bound body at the ISCO of a supermassive black hole — orbits at locally measured speed $c/2$, and that is the velocity scale channel 3 can imprint on its ejecta in a single exchange. [I — for this channel, per encounter; the chain question is §4.3]

Spin. For a maximally spinning Kerr hole ($a^* \rightarrow 1$) the prograde ISCO descends toward GM/c^2 and orbital speeds rise; astrophysical spins are bounded away from extremality (the Thorne limit $a^* \approx 0.998$), and we grant $0.6\text{--}0.7c$ as the heroic ceiling. **The Penrose loophole closes itself:** extracting the hole's rotational energy to eject a fragment faster requires the body to split inside the ergosphere with fragment separation velocity exceeding $c/2$ (Wald 1974) — that is, it requires a relativistic launcher *already operating* at the very speed scale in question, embedded in the payload; the "gravitational" pathway then contains a non-gravitational engine and inherits every dissipative inequality of §§5–8. Passive dynamics gets no Penrose bonus. Even at the heroic ceiling:

$$\gamma(0.5c) = 1.155 \rightarrow E_k = 0.155 \text{ mc}^2; \quad \gamma(0.7c) = 1.400 \rightarrow E_k = 0.400 \text{ mc}^2.$$

Against the requirement $\gamma = 7.09$, $E_k = 6.09 \text{ mc}^2$, the shortfall is a factor of **39 in energy** at $0.5c$ (15 at $0.7c$). No configuration of gravitational dynamics ejects a passive strength-bound body at $\gamma = 7$. [I]

4.3 Chained encounters: the kinematics permit, the mechanism refuses

Correction owned first. An earlier draft claimed "chains asymptote; they do not diverge." That is wrong as kinematics: relativistic boosts compound in *rapidity*, which is additive — $w(0.5c) = 0.549$, $w(0.99c) = 2.647$, so five ideal $0.5c$ boosts, each delivered in the body's instantaneous frame, exceed $0.99c$. If the universe offered a relay of ideal slingshot stations, the chain would work. The closure is not kinematic; it is mechanical, and it is sharp:

The gain mechanism self-destructs after the first boost. Channel 3 — the only channel that manufactures $\sim 0.5c$ -scale gains — is an *exchange*: the body must be captured into, or arrive bound within, a system that then ejects it. Exchange requires the body to approach the system at v_∞ small compared to the system's internal

orbital velocities (capture cross-sections collapse as v_{∞} grows; a body incident at $v_{\infty} \geq 0.5c$ on *any* bound system is in the elastic-flyby regime — channel 1 — and leaves with its speed unchanged). So after boost one, the body arrives at every subsequent system at $\geq 0.5c$ relative speed, and no subsequent exchange can grip it. Boost two never happens by channel 3. Channel 2 remains available at $\geq 0.5c$ — and adds $\leq 0.03c$ per pass against the measured black-hole velocity distribution, needing dozens of aligned, maximally kicked merger remnants to matter, each pass re-running the §5 environment. [I for the per-mechanism caps (energy conservation; measured velocity distribution; capture-cross-section collapse); P for any contrived relay]

This is a stronger closure than the one it replaces, but it is honestly *mechanism-shaped*, not a single inequality: it says every known gravitational gain mechanism is either elastic, capped by source velocities, or requires an approach speed the previous boost has already destroyed. A referee who exhibits a gravitational gain mechanism outside channels 1–4 reopens this section; we know of none.

4.4 Below the ceiling: the sub-0.5c gravitational fluke [P]

Below $\sim 0.5c$ the pathway exists in principle. It is rate-suppressed to effective nonexistence by the conjunction of: (a) capture phase space — deep survivable passes require quiescent compact objects, while rocks form in metal-rich stellar environments, not black-hole merger neighborhoods, and gravitational-wave capture cross-sections are minuscule; (b) environmental dose — accreting systems bathe the pericenter region in radiation that vaporizes small bodies (an SMBH at even 10^{-4} of Eddington outshines the Sun by $\sim 10^6$ within the pericenter distances at issue), while quiescent systems are the rare dark subset; (c) post-ejection erosion in the interstellar medium (§5 applies from $0.1c$ upward with force). Each factor is a probability statement, and we tag the conjunction [P]: *no inequality forbids a $0.3c$ pebble; the universe simply has almost no machinery for making one and active machinery for destroying it in transit.*

Conclusion of Case G: gravity, the unique heat-free coupling, has no known or physically credible channel to $\gamma = 7$ for strength-bound solids — every enumerated mechanism is elastic, capped by measured source velocities, capped at $\sim 0.5c$ per exchange, or requires an approach speed the previous boost has already destroyed. This is a mechanism-by-mechanism closure resting on dynamics, not an observational absence — but it is not a single-inequality theorem, and we do not call it one. [I per channel; P on relays]

5. Case M-environmental: ordinary interstellar matter is a particle beam [I at $0.99c$, parameterized by column]

A body at $\beta = 0.99$ moving through interstellar gas of density n intercepts, in its own frame, a hadron beam of energy $(\gamma-1)m_p c^2 = \mathbf{5.7 \text{ GeV per hydrogen atom}}$ at number flux $\gamma n \beta c$. This case is propulsion-independent — it binds beamed sails, rockets, mass drivers, and naturally flung rocks equally, *wherever they fly through ordinary interstellar matter*. It is a transport constraint, not an acceleration constraint: an engineered vacuum corridor or a rarefied-phase trajectory evades it, and those cases are owned in §9.5, priced at what evading it costs. Everything in this section is parameterized by density n and column $N = \int n \, dl$, so the reader can move the environment and watch the numbers follow.

5.1 Frontal heating of a thick body

For any body thicker than the hadronic cascade depth (nuclear interaction length $\sim 66 \text{ g/cm}^2$ in rock, $\sim 25 \text{ cm}$; cascades deposit the bulk of 6-GeV energy within $\sim 1 \text{ m}$), the intercepted power per unit frontal area at ISM density n is

$$P = \gamma n \beta c (\gamma-1) m_p c^2.$$

At the mean warm-ISM density $n = 1 \text{ cm}^{-3}$ and $\beta = 0.99$: $P = 1.9 \times 10^6 \text{ W/m}^2$. A body radiating from front and back faces equilibrates at

$T_{eq} = (P/2\sigma)^{1/4} = 2,030 \text{ K}$ (2,410 K if only the leading face radiates effectively).

$n \text{ (cm}^{-3}\text{)}$	$P \text{ (W/m}^2\text{)}$	$T_{eq} \text{ (K, 2-face)}$
0.1	1.9×10^5	1,140
1	1.9×10^6	2,030
10	1.9×10^7	3,610
100	1.9×10^8	6,420

Silicates melt at 1,400–1,900 K; iron at 1,811 K; tungsten at 3,695 K; graphite sublimates near 3,900 K; the most refractory known carbides fail by ~4,300 K. **At ordinary disk densities ($n \geq 0.3 \text{ cm}^{-3}$), $\beta = 0.99$ melts every ordinary rock and every structural metal; any region denser than $\sim 20 \text{ cm}^{-3}$ exceeds the failure temperature of every solid known.** [I, at the stated densities]

The low-density escape, priced. The thermal kill is genuinely density-dependent: at $n = 0.01 \text{ cm}^{-3}$ (hot ionized bubbles) the load drops to ~19 kW/m² and $T_{eq} \approx 640 \text{ K}$ — thermally survivable; in the halo or intergalactic space ($n \sim 10^{-5} \text{ cm}^{-3}$) it is ~19 W/m² and ~114 K — thermally trivial. What low density does *not* buy: the damage channels below are charged per unit *column*, and a trajectory that begins or ends anywhere interesting — any stellar system, including the origin and destination of every conceivable mission — accretes disk-phase column at both ends; the Local-Bubble-class cavities are ~10² pc wide in a 10⁴-pc disk. A body that stays forever in the halo evades this section and has nowhere to have come from and nowhere to arrive; §9.5 owns it. [I at stated n ; the routing question is [P/E]]

The survivors of the $n = 1 \text{ cm}^{-3}$ row (tungsten, graphite, ultra-refractory carbides at their limiting temperatures) are not rocks but curated monoliths, and they do not escape: at 2,000–2,400 K they sublime continuously in vacuum; every square meter of frontal surface absorbs a fluence of 2.2×10^{24} six-GeV protons per 10 parsecs (an integrated $2 \times 10^{15} \text{ J/m}^2$, delivered as hadronic cascades that amorphize, embrittle, and spall the lattice); and the accompanying dust — ~1% of ISM mass in ~0.1 μm grains — arrives as 4.6 J point detonations (each grain carries the energy of a rifle round in a cross-section of 10^{-14} m^2) at ~10¹¹ impacts per m² per year, cratering the face faster than sputtering alone. Deceleration is symmetric: any mission profile that ends with intact matter must run this gauntlet twice. [I for heating]

The erosion rates have a quantitative literature anchor at lower β . Hoang, Lazarian, Burkhart & Loeb (2017) computed, for quartz surfaces at $v = 0.2c$: gas bombardment (including heavy-ion track formation) damages a surface layer ~0.1 mm deep over a column $N_H \approx 2 \times 10^{18} \text{ cm}^{-2}$ (Karlušić 2017 argues the track model overestimates this; we carry it as an upper anchor), while **dust cratering by explosive evaporation erodes ~0.5 mm over $N_H \approx 3 \times 10^{17} \text{ cm}^{-2}$** — roughly 30× more erosive per unit column than gas, and, at the mean density $n = 1 \text{ cm}^{-3}$ ($3.1 \times 10^{18} \text{ cm}^{-2}$ per parsec), **~5 mm of surface per parsec traveled, already at 0.2c**. Drobny et al. (2021) add a subsurface channel that survives even perfect erosion mitigation: implanted H/He accumulates below the surface (Bethe–Bloch range) and drives blistering and exfoliation. At 0.99c every impactor carries 30× the 0.2c energy (5.7 GeV vs. 19 MeV per nucleon) and the thermal channel of §5.1 operates on top. Centimeters of frontal loss per tens of parsecs is the *floor*; the melting inequality above is the ceiling. [I at 0.99c via heating; erosion rates cited at 0.2c and extrapolated — the extrapolation is ours and is conservative in direction, since per-impact energy and cascade depth both grow.]

5.2 Thin bodies do not escape by transparency

A 1 g/m² sail is $\sim 10^{-4}$ g/cm² thick; a 6-GeV proton deposits only ~ 200 eV traversing it (minimum-ionizing $dE/dx \approx 2$ MeV per g/cm²), for a deposited power of ~ 0.07 W/m² — thermally negligible. But the sail pays instead through the damage channel: every traversing proton leaves an ionization track, every ~ 0.1 μ m dust grain removes a crater orders of magnitude larger than itself in a film with no bulk to spare, and over interstellar fluences (10^{24} p/m² per 10 pc) the accumulated displacement damage in a membrane a few hundred atoms thick destroys the optical properties (α , ϵ , R) on which the sail's thermal survival depends (§6–7). A thin body trades the heating death for the erosion death. [I on fluence; erosion-rate constants from Hoang et al. 2017]

Nature has already run the $L \rightarrow 0$ limit of this experiment. Hoang, Lazarian & Schlickeiser (2015), examining the old proposal (Spitzer 1949; Hayakawa 1972) that relativistic dust grains are ultra-high-energy cosmic-ray primaries, found that sub-micron solid grains (a) cannot be radiation-pressure accelerated beyond $\gamma < 2$ even by the most powerful astrophysical sources, and (b) if somehow made relativistic, are destroyed — Coulomb-exploded by collisional charging after sweeping a gas column of only $\sim 10^{17}$ cm⁻² (**0.03 parsecs** at $n = 1$ cm⁻³). The smallest condensed solids fail *faster* than macroscopic ones (charging stress is a surface effect); the mesoscale is closed at both ends. [I, published]

5.3 Passive shielding is self-defeating

Stopping the cascade requires ~ 5 nuclear interaction lengths: ~ 430 g/cm² = **4,300 kg/m²** of graphite. For the beamed-sail pathway, whose entire scheme rests on areal densities of order 1 g/m² (§7), a shield costs **4×10^6 times the mass budget** — the shield requirement and the acceleration requirement are contradictory by six orders of magnitude. For a thick monolith the shield *is* the melting face of §5.1. For passive matter there is no third option: it either stops the beam (and takes its power) or passes it (and takes its damage). [I]

5.4 Active shielding, named and priced

A referee will ask about deflecting the beam instead of stopping it, so we enumerate. A 5.7-GeV proton has magnetic rigidity 6.6 GV: gyroradius ~ 22 m at 1 T, ~ 2.2 m at 10 T. Deflecting the *ionized gas* component with a superconducting forward coil is therefore not dimensionally absurd — and it fails to save the pathway for reasons that are each independently sufficient:

1. **The warm ISM is largely neutral.** Neutral hydrogen ignores the field until ionized; an ionizing standoff system (laser or electron curtain) must process the full oncoming column, adding its own power, mass, and waste heat to a budget already fatal.
2. **Dust is magnetically immune and dominates the damage.** Grain charge-to-mass ratios are minuscule at these rigidities; the $\sim 30\times$ -per-column erosion channel of §5.1 (Hoang et al. 2017) passes straight through any field. Active shielding addresses the lesser channel.
3. **Momentum still arrives.** Deflection transfers the beam's momentum to the coils: the drag power is not removed, only relocated from ablation to structural load and coil heating.
4. **The shield lives in the flux it deflects.** Superconductors under GeV-hadron and photon bombardment quench and transmute; coil mass, cryogenics, and radiation shielding for the shield re-enter the §5.3 mass spiral.

Active shields may reduce the *ionized-gas dose*; they do not touch dust, neutrals, drag, or their own mass problem. [I on channels 2–3; E on the engineered system as a whole]

6. Case EM- γ , part 1: the 14 $\times$ Doppler burden is conserved, and no material class carries it [I over measured classes]

The beamed sail earned its counterexample status (§2.2) with three simultaneously held properties: $\alpha \sim 10^{-6}$, areal density $\sim 1 \text{ g/m}^2$, high reflectivity R . All three are demonstrated — *separately, at fixed wavelength, at centimeter scale*. The acceleration then imposes a bandwidth burden that must land somewhere in the photon chain.

6.1 The 14× factor, and why chirping relocates it rather than removing it

As the sail accelerates $0 \rightarrow 0.99c$, the Doppler factor between transmitter frame and sail frame grows by

$$D(\beta) = \sqrt{(1+\beta)/(1-\beta)} = 14.1 \text{ at } \beta = 0.99.$$

For a **fixed-frequency** transmitter, the sail-frame wavelength sweeps $\lambda_0 \rightarrow 14\lambda_0$: the mirror must hold its absorption floor over three-plus octaves.

An earlier draft treated that sweep as unavoidable. It is not: a **chirped transmitter** can sweep its emission frequency *upward* by the same factor so the sail-frame wavelength stays fixed at the mirror's design band. Nothing in physics forbids this. What it does is relocate the 14× burden onto the source side, where the endpoint arithmetic is unforgiving: to hold a sail-frame band at $1 \text{ }\mu\text{m}$, the transmitter must finish the burn emitting at $1,000/14.1 = 71 \text{ nm}$ — **vacuum ultraviolet, 17.5 eV photons**, above the bandgap of every optical material. There are no low-loss mirrors, windows, or phase-control optics below the LiF cutoff ($\sim 105 \text{ nm}$); the best VUV reflectors sit at 20–40% reflectivity, i.e. $\alpha \sim 0.6$, nine orders of magnitude off the loss floors a petawatt-class beam train requires; and no coherent source technology delivers even gigawatts, let alone the $\sim 10^{14} \text{ W/kg}$ of §7, anywhere near that band. Splitting the chirp (source sweeps $k\times$, sail sweeps $14/k\times$) only divides the burden; the product is fixed by kinematics.

The correct statement is therefore a conservation law: the photon chain as a whole must span a 14× band at the loss floor its thermal budget demands — on the sail (§6.2–6.3), on the transmitter optics, or split between them — and no measured material class carries it at any point in the chain. [I over measured classes at both ends; E against undiscovered source/optics technology, said plainly]

6.2 Why ppm floors are narrow-band physics

Demonstrated $\alpha \sim 10^{-6}$ mirrors are quarter-wave dielectric stacks: interference structures whose reflectivity band is set by the index contrast, with fractional bandwidth $\Delta\lambda/\lambda \sim 0.2\text{--}0.5$. Outside the stopband, reflectivity collapses and the field penetrates the full stack: absorption rises by 3–6 orders of magnitude. No interference structure holds a stopband over a 14× ratio: a chirped or stacked-stack design covering three octaves multiplies layer count and thickness — and the absorption of a dielectric stack scales with the material the field samples, so broadening the band raises areal mass and α together. Worse, the sweep runs from $1 \text{ }\mu\text{m}$ *into the mid-infrared*, where every oxide and nitride has its reststrahlen/phonon bands (SiO_2 absorbs catastrophically at $8\text{--}9.5 \text{ }\mu\text{m}$; a $1\text{-}\mu\text{m}$ -designed stack swept to $14 \text{ }\mu\text{m}$ crosses its own lattice resonances, $\alpha \rightarrow 10^{-1}$). Metals are genuinely broadband — and improve toward the infrared (Hagen–Rubens, $\alpha \propto \sqrt{\nu}$) — but their floor across the band is $\alpha \sim 10^{-3}\text{--}10^{-2}$ (finite Drude conductivity), three to four orders above budget. Sweeping α from 10^{-6} to even 10^{-4} multiplies the absorbed dose of §2.2 by 100; the thermal budget that closed in hours now requires the sail to radiate at powers implying equilibrium temperatures far beyond any coating's damage threshold (§7 quantifies).

Tagging this claim honestly: we searched for, and the literature does not provide, a first-principles theorem bounding the absorption of an arbitrary reflector over a fixed band (the Kramers–Kronig relations couple dispersion to loss and forbid a *lossless, massless, broadband* mirror as a consistent set of optical constants, and Drude conductivity floors metals, but no published inequality covers all conceivable structures). The claim is therefore tagged: **[I over every measured mirror class]** — dielectric stacks, photonic crystals, metals, across thousands of published absorption spectra, none sustains $\alpha \leq 10^{-5}$ over a 14× band, and the mid-infrared

phonon physics the sweep runs into points the wrong way — **and [E] against undiscovered metamaterials**. A referee who exhibits a ppm-absorption three-octave gram-scale reflector falsifies this tier; none exists in the record.

6.4 The defect-statistics wall

All ppm absorptivities on record are measured on centimeter-scale superpolished substrates. A sail is 10^6 – 10^9 times that area, and its survival at $\sim 10^{11}$ W/m² of continuous illumination is governed not by mean absorption but by the extreme-value statistics of the *worst* defect on the sheet: a single absorbing inclusion is a burn-through site, and burn-through in a stressed membrane under petawatt illumination is not a local event. Large-area coating at ppm uniformity has never been demonstrated, and area scaling of damage thresholds is a known, adverse, extreme-value regime — the mean is not the operative statistic. [E — a demonstration gap, tagged as such, sitting *underneath* the [I]-level closures of §§6.2–6.3.]

6.3 Emissivity closes the pincer from the other side

The sail's only cooling channel is thermal radiation, and a film optically thin at 1 μm is optically thin — hence a *bad emitter* — at its own 3–10 μm emission wavelengths: real sub-micron films have hemispherical emissivities $\epsilon \sim 0.01$ – 0.1 unless deliberately structured (which adds mass and its own absorption). Equilibrium temperature scales as $\epsilon^{(-1/4)}$. Rerunning §2.2's closed budget with honest parameters:

α (swept)	ϵ	T_eq for the baseline burn
10^{-6}	1.0	875 K — closes (the §2.2 idealization)
10^{-6}	0.1	1,556 K — marginal
10^{-6}	0.05	1,851 K — coating death
10^{-4}	0.05	> 5,800 K nominal — vaporized long before

The α that survives a 14× sweep and the ϵ of a real gram-per-square-meter film cannot coexist with any coating's survival temperature. **The two requirements — reflect a three-octave band at ppm absorption, and radiate the residue from a sub-micron film — are jointly unsatisfiable in measured material physics.** [I over all measured classes; formally [E] against undiscovered metamaterials, and we say so plainly.]

7. Case EM-γ, part 2: the thermal–diffraction scissors [I/E]

Independently of coatings, the beam must *stay on the sail*, and here two timescales cross.

Thermal floor on burn time. Absorbed energy αE_{inc} must be radiated at $\leq 2\epsilon\sigma T^4$ per unit sail area ($T \leq$ coating limit):

$$t_{\text{burn}} \geq t_{\text{min}} = \alpha E_{\text{inc}} \rho_A / (2\epsilon\sigma T^4), \quad \rho_A = \text{areal density}.$$

Diffraction ceiling on burn time. A transmitter of aperture D at wavelength λ holds its spot within a sail of size s only out to $d_{\text{max}} \approx sD/(2.44\lambda)$; the sail, spending the burn near c , exits the focus after

$$t_{\text{burn}} \leq t_{\text{max}} = s D / (2.44 \lambda \langle \beta \rangle c).$$

The scissors condition $t_{\text{min}} \leq t_{\text{max}}$ yields the required aperture:

$$D \geq 2.44 \lambda \langle \beta \rangle c \cdot \alpha E_{\text{inc}} \rho_A / (2\epsilon \sigma T^4 s) .$$

Sweeping the parameter space:

α	ϵ	T (K)	s (m)	D required
10^{-6}	1.0	1,000	30	117 km
10^{-6}	0.5	1,500	30	46 km
10^{-5}	0.1	1,500	100	695 km
10^{-4}	0.05	1,000	30	2.3×10^5 km (18 Earth diameters)

Read the first row honestly: with the idealized (§2.2) sail, the aperture is "only" ~100 km of optically coherent, phase-locked emitter — pushing $\sim 1.1 \times 10^{14}$ W per kilogram of payload (a 100-kg body needs ~11 PW, five hundred times humanity's mean power output, held for over an hour at micro-radian pointing across ~10 AU), while the sail rides at ~5,800 g mean acceleration (a bare wafer survives, machinery does not). No physical law forbids a 100-km laser; we tag the row [E] and place it in the boundary cases. But the idealized row is not available: §6 showed the 14× bandwidth burden must land on the sail or the transmitter chain (neither carries it), and ϵ cannot reach 1 in a gram-scale film. With honest optical constants the aperture inflates to planetary diameters and the beam power to stellar fractions — and §5 has already charged the sail $4 \times 10^6 \times$ its mass budget for ISM shielding it cannot buy, then destroys its coating in transit anyway. **The conjunction closes: every point of the (α , ϵ , T, s, ρ_A) space fails at least one of — Doppler-band absorption (§6.2), emissivity (§6.3), diffraction aperture (§7), ISM (§5).** [I over measured physics, with the [E] boundary row owned in §9.]

Slowing down does not help: a longer, gentler burn covers more distance, and the aperture requirement *grows* with burn distance. The scissors cut in both directions; this is why the eon-timescale boundary case (§9.1) must abandon beaming altogether.

8. Cases M-rocket, EM-s, C: onboard and structured-force propulsion [I]

8.1 The rocket inequality

The relativistic rocket equation for exhaust speed w gives the mass ratio

$$m_0/m_1 = [(1+\beta) / (1-\beta)]^{(c/2w)} .$$

β	chemical ($w = 1.5 \times 10^{-5}c \approx 4.5$ km/s)	fission fragments (0.05c)	fusion (0.1c)	photon (c)
0.10	10^{2905}	7.4	2.7	1.11
0.50	10^{15904}	5.9×10^4	243	1.73
0.90	10^{42625}	6.1×10^{12}	2.5×10^6	4.36
0.99	10^{76628}	$\sim 10^{23}$	3.1×10^{11}	14.1

(An earlier draft's chemical column used $w = 5 \times 10^{-5}c$ while labeling it chemical; both header and numbers now use real chemical exhaust, 4.5 km/s. The conclusion was never in doubt at either value.) Chemical and fission propulsion are dead at any relativistic β — mass ratios whose *exponents* exceed the particle count of the observable universe. Fusion at 0.99c demands a mass ratio of 3×10^{11} : three hundred billion kilograms of fuel

— a ~600-m asteroid — per kilogram delivered, before a single inefficiency is charged; a Ceres-mass vehicle (9.4×10^{20} kg) would deliver about 3×10^9 kg. [I]

8.2 The waste-heat inequality

Any onboard converter of efficiency η handling the energy flow leaves $W = E_k(1-\eta)/\eta$ in the ship. At $\beta = 0.99$ and a generous $\eta = 0.8$: $W = 1.37 \times 10^{17}$ J/kg. Radiating at a space-reactor-class 200 W per kilogram of ship: **2.2×10^7 years**. At a fantastical 1 kW/kg and $\eta = 0.99$: 175,000 years — during which the "burn" is limited by its own radiator, and the ship crosses the §5 gauntlet at increasing β with its radiators forward. The photon rocket avoids reaction-mass arithmetic (mass ratio 14.1) but must generate and *emit as light* $7.0 \text{ mc}^2 = 6.3 \times 10^{17}$ J per delivered kilogram through an onboard engine whose every percent of inefficiency is 10^{15} J/kg of onboard heat: it is the waste-heat inequality in its purest form. Onboard pathways fail on the conjunction (mass ratio) $\wedge$ (waste heat): each is independently fatal at 0.99c. [I]

8.3 Structured-force machines (mass drivers, coilguns, charged accelerators)

Any external machine gripping the body through electromagnetic structure transmits force through the body's own material: acceleration is capped by strength,

$$a \leq S/(\rho L) \approx 3.3 \times 10^4 \text{ m/s}^2 \approx 3,400 \text{ g} \quad (1\text{-m body at } S = 10^8 \text{ Pa, } \rho = 3,000),$$

so the track length obeys

$$\ell \geq (\gamma-1)c^2\rho L/S \approx 1.6 \times 10^{13} \text{ m} \approx 110 \text{ AU} \quad (\text{at 100\% of ultimate strength; } \sim 1,100 \text{ AU at a sane working stress}).$$

An accelerator a hundred to a thousand AU long, aligned and powered to deliver 5.5×10^{17} J/kg with the payload's dissipative response (eddy currents in any conductor under changing fields, hysteresis, flux penetration in superconductors driven at their critical limits) charging heat *into the payload* throughout — and the exit door opens onto §5. No single inequality kills the idealized frictionless version; the conjunction (track scale [E]) $\wedge$ (payload dissipation floor [I-material]) $\wedge$ (ISM at exit [I]) does. The body also cannot be gripped gravitationally to evade the strength cap — that is Case G, already closed. [I $\wedge$ E]

9. Boundary cases, owned in the main text

The argument's honesty rests on stating exactly what it does not forbid.

9.1 Eon-timescale acceleration. [Open in principle] Nothing above forbids accelerating a body gently for geological time: at a sustained 10^{-3} m/s² proper acceleration, 0.99c arrives after ~70,000 years — and $\sim 5 \times 10^{20}$ m of path, **eighteen kiloparsecs, a galactic radius**. The heat-rejection inequalities all relax as $t_{\text{burn}} \rightarrow \infty$; everything charged per unit *distance* does not. Beamed schemes must hold the §7 focus over that whole path — apertures beyond planetary scale, the scissors again; onboard schemes still pay §8.1's mass ratios in full (mass ratio is time-independent); the §5 ISM dose is charged per parsec, and an 18-kpc run at climbing β through unchosen interstellar weather is §5 compounded thousands of times over; and any machine holding itself together for 10^5 years exceeds every engineering-lifetime datum by orders of magnitude. We class this case [E]: not forbidden by inequality, closed for any bounded actor, and — decisively — the ISM and Doppler constraints it cannot relax still bind at the destination velocity.

9.2 The sub-0.5c gravitational fluke. [P] As §4.4: a strength-bound pebble ejected at 0.1–0.5c by a deep encounter with a quiescent massive black hole violates no inequality. Its production channels are rate-

suppressed (phase space, environment), and §5 erodes it in transit (a 0.3c pebble intercepts $\sim 7 \text{ kW/m}^2$ in a 10 cm^{-3} cloud — thermally survivable at $\sim 500 \text{ K}$ equilibrium, but cratering- and sputtering-limited on kiloparsec paths at Hoang-et-al. erosion rates). We bound, we do not forbid.

9.3 The idealized single-wavelength wafer. [E] The §7 first row: a gram-scale bare wafer, a 100-km coherent aperture, petawatts, and a trajectory short enough to out-run coating damage. Every added requirement (a payload; deceleration; arrival intact through §5; the Doppler sweep it cannot actually evade) re-closes it. It survives as a reminder that the claim is about *macroscopic intact matter*, not about whether photons push.

9.4 What the claim never covered. Particles and plasma reach $\gamma \gg 7$ daily, in nature (cosmic rays to 10^{20} eV) and in machines. The argument is precisely about the gap between them and condensed matter: binding energies of eV per atom versus kinetic demands of GeV per nucleon — a nine-order-of-magnitude mismatch between what holds a solid together and what the journey delivers to every atom of its leading face.

9.5 Engineered vacua and rarefied-phase trajectories. [E/P] The §5 tier is environmental, so environments evade it. (a) *An artificial vacuum corridor* — an evacuated accelerator long enough for the burn — is §8.3's machine: $\geq 110 \text{ AU}$ of track at ultimate material strength ($\sim 1,100 \text{ AU}$ at working stress) for a meter-scale payload, plus the payload-side dissipation floor, plus the exit into whatever medium lies beyond. Known physics does not forbid it; we price it and tag it [E]. (b) *A trajectory confined to rarefied phases* (hot bubbles, galactic halo, intergalactic space) is thermally survivable at 0.99c (§5.1: $\sim 114 \text{ K}$ at halo densities). But acceleration still requires one of the closed pathways of §§6–8; and a payload that both originates and arrives at anything — any star system, any planet — buys disk-phase column at both ends of the journey. The rarefied-phase case is a real opening for a body with no origin and no destination; for transport between places where matter actually resides, it reduces to a partial mitigation of §5, not an escape. [P on routing; E on the acceleration it still needs]

10. Prior art

In a targeted literature search (fanned across the propulsion, astrophysics, materials, and thermodynamics corpora, with adversarial verification of every citation below against its primary source) **we found no prior source that assembles these constraints into a coupling-class impossibility argument for ordinary macroscopic condensed bodies at $\beta \approx 0.99$** . This is a negative claim held at the confidence a negative deserves — the search was broad, not exhaustive, and one obscure paper would revise this section, not the physics. The components have distinguished pedigrees; we claim novelty for the assembly, the coupling-classification closure, and the channel-by-channel gravitational analysis for strength-bound bodies.

- **The closest prior art: Semyonov.** Oleg G. Semyonov, "Relativistic rocket: Dream and reality" (Acta Astronautica 99, 2014, 52–70) and "Pros and cons of relativistic interstellar flight" (Acta Astronautica 151, 2018, 736–742; arXiv:1807.08608) identifies exactly two physical factors "that stand against our dream of the stars — thermodynamics and radiation hazard": the onboard power generator and heat-disposing radiator grow the dry mass and cap the achievable speed (our §8.2), and ship-frame interstellar gas "turns into oncoming flux of hard ionizing radiation" while relativistic dust causes mechanical damage (our §5). He is the nearest published position to this paper's — and he explicitly stops short of it, proposing antimatter fuel, shielding, and beamed-matter propulsion as mitigations rather than concluding impossibility. Where he offers engineering hope, our §§5–8 offer the closing inequalities.
- **A published formal ceiling — in a different regime.** Yurtsever & Wilkinson, "Limits and signatures of relativistic spaceflight" (Acta Astronautica 142, 2018, 37–44; arXiv:1503.05845) prove the vacuum itself imposes a sub-light ceiling on any macroscopic baryonic object: blueshifted CMB photons exceed the e^+e^- pair-production threshold when $\gamma(1+\beta) \geq 2.47 \times 10^8$, i.e. $\gamma \sim 10^8$, with drag and ionization channels opening from $\gamma(1+\beta) \sim 10^4$. This is the only formal no-go-style theorem we find in print, and it is

complementary to ours: at $\gamma = 7.09$ the CMB blueshifts to a harmless $\sim 10^{-2}$ eV (which is why the CMB appears nowhere in our tiers). Their theorem caps the far ultrarelativistic regime from the environment side; ours closes the 0.99c regime from the coupling-and-materials side.

- **Beamed propulsion, by its own advocates.** Marx (Nature 211, 1966) proposed laser propulsion to relativistic speeds; Redding (Nature 213, 1967) identified the deceleration problem at once. Forward (J. Spacecraft & Rockets 21, 1984) developed the staged lightsail. Lubin, "A Roadmap to Interstellar Flight" (JBIS 69, 2016; arXiv:1604.01356) is the maximal contemporary case for directed energy, and its own numbers are the concession: a full DE-STAR 4 — a 50–70 GW, kilometer-scale phased array — propels a *gram-scale wafer* with a 1 m sail to $\sim 0.26c$; the same system takes a 100 kg payload only to $\sim 0.01c$ and 10^5 kg to $\sim 10^3$ km/s. Kulkarni, Lubin & Zhang (AJ 155, 155, 2018; arXiv:1710.10732) confirm diffraction caps gram-scale craft near 0.2c. The steep mass–velocity cliff in the advocates' own roadmap is §7's scissors, observed from inside. The Breakthrough Starshot materials literature (Atwater et al., Nature Materials 17, 2018; Ilic, Went & Atwater, Nano Letters 18, 2018) states the reflectivity–mass–absorption–emissivity trades of §6 as the central unsolved sail problem — at 0.2c, where the Doppler sweep is 1.22× rather than our 14×.
- **ISM damage.** Hoang, Lazarian, Burkhart & Loeb (ApJ 837, 5, 2017; arXiv:1608.05284) give the canonical quantitative treatment at 0.2c: gas (with heavy-ion track formation) damages quartz to ~ 0.1 mm depth over $N_H \sim 2 \times 10^{18} \text{ cm}^{-2}$ (Karlušić, arXiv:1701.04319, argues this overestimates track damage — carried here as a qualification); dust cratering erodes ~ 0.5 mm over $N_H \sim 3 \times 10^{17} \text{ cm}^{-2}$, $\sim 30\times$ more erosive per column. Antecedents: Early & London (2000, 2018) on lightsail dust damage. Drobny et al. (ApJ 908, 248, 2021; arXiv:2103.07517) and Cohen et al. (ApJ 930, 74, 2022) add subsurface gas-accumulation blistering. All frame damage at 0.1–0.3c as an engineering challenge; our §5 extends the same mechanism inventory to $\beta = 0.99$, where per-nucleon energy crosses into the GeV hadronic-cascade regime and the *thermal* channel becomes independently fatal — that extension is ours, and it converts a challenge into an inequality.
- **The natural analog.** Hoang, Lazarian & Schlickeiser (ApJ 806, 255, 2015; arXiv:1412.0578): sub-micron grains cap at $\gamma < 2$ under radiation-pressure acceleration by the most powerful sources known, and relativistic grains are Coulomb-exploded after $\sim 10^{17} \text{ cm}^{-2}$ of transit — effectively retiring the Spitzer (1949)/Hayakawa (1972) relativistic-dust cosmic-ray hypothesis. Published, peer-reviewed non-survivability of condensed matter at relativistic speed, in the $L \rightarrow 0$ limit.
- **Gravitational ejection ceilings.** Hills (Nature 331, 687, 1988) founded binary-disruption ejection ($\sim 4 \times 10^3$ km/s). Guillochon & Loeb, "The fastest unbound stars in the universe" (ApJ 806, 124, 2015; arXiv:1411.5022) find the classical mechanism caps near 10^4 km/s (0.03c) and that stars bound to the secondary of a merging massive-black-hole binary are occasionally ejected at $\sim 10^5$ km/s — $c/3$ — with the ceiling set by tidal disruption at the deepest survivable pericenter. The observed record: S5-HVS1, 1,755 km/s (Koposov et al., MNRAS 491, 2020). Our §4 transposes their tidal-limit logic from self-gravity-bound stars to strength-bound solids, which survive deeper, and finds the mass-independent per-exchange scale $c/2$ ($\lesssim 0.7c$ Kerr) — then closes the chain question mechanically (exchange capture collapses above the first boost's exit speed). We found no prior statement of this analysis for strength-bound bodies; it is the argument's decisive step for the gravitational branch.
- **Thermodynamic bounds: the theorem that is not there.** A universal entropy-production lower bound for "any coupling that accelerates a body" would upgrade Lemma 2 from classification to derivation. Our search did not find one, and the structure of the finite-time-thermodynamics literature explains why: its bounds (optimal-transport dissipation bounds, Aurell et al. 2011; the thermodynamic-speed-limit family) price *state-space* change per unit time and vanish in the slow-driving limit, while quantum speed limits (Margolus–Levitin type) are non-binding by tens of orders of magnitude at macroscopic energies. **We therefore do not claim the original proposal's Hypothesis 3 ("every coupling has finite irreversibility") as a theorem.** The heat floors in this paper come from measured material response — α , ϵ , conductivity — not from a general principle; and the one coupling with no material response,

gravity, is exactly the one we close dynamically, mechanism by mechanism. Hypothesis 3 survives only in the amended, case-checked form: *every coupling either deposits heat in the payload through its measured material response, or lacks a credible production channel beyond $\sim 0.5c$ per encounter.*

The assembly is the argument. Each literature above closes its own corridor and stops. Semyonov closes rockets and gestures at sails; Lubin champions sails and concedes grams; Hoang et al. erode sails and stop at $0.3c$; Guillochon & Loeb cap gravity and consider only stars. Assembled, with the coupling classification of §3 guaranteeing no unexamined corridor remains, they are no longer a collection of engineering complaints but a closed case analysis. That closure — and the channel-by-channel analysis that prices the one heat-free door — is what this paper adds.

11. Discussion: nature's record, read correctly

The corroboration (never a premise): nature runs every accelerator we cannot. Supernovae, kilonovae, SMBH slingshots, pulsar winds, jets — 13.8 Gyr of them, across $\sim 10^{11}$ galaxies. The census of what these deliver at relativistic speed is unanimous: **particles and plasma**. Kilonova ejecta at $0.1\text{--}0.3c$ arrive as r-process plasma. Jets at $\gamma \sim 10\text{--}50$ are ionized flows. Cosmic rays reach macroscopic *energies* (10^{20} eV) in single nuclei. Meanwhile the record for observed intact-body speed stands at S5-HVS1's $1,755\text{ km/s} \approx 0.006c$ (a star; S4714's pericenter dash at $\sim 0.08c$ is a momentary orbital speed, also a star); the theoretical stellar ceiling is Guillochon & Loeb's occasional $c/3$; and for solids the record-holders are the three known interstellar objects at $26\text{--}60\text{ km/s}$ — $10^{-4} c$. The argument explains this record; the record did not produce the argument. We assign the non-observation of fast rocks zero weight (a dark meter rock at $0.5c$ has detection probability ≈ 0 in every existing survey: expected detections ≈ 0 under either hypothesis) — and note that the *positive* record, of what relativistic accelerators demonstrably emit, is Bayesian corroboration of exactly the dichotomy the argument establishes: nature's mechanisms all couple through the channels of Lemma 1, and everything they fling fast, they first tear apart.

The claim's modality bears final restatement. We have not proven a law of nature, and we no longer call this a theorem in the unconditional sense. What we have exhibited is that intact macroscopic condensed matter at $\beta = 0.99$ is trapped between three killers — the energy scale (6.09 rest-masses per kilogram), coupling and material dissipation, and relativistic environmental impact — such that gravity avoids the heat but has no credible $\gamma \approx 7$ production channel, and every engineered alternative moves its contradiction into absurd artifact scale, thermal rejection, shielding mass, or ISM damage, with each closure tagged at exactly the strength it earns. New physics could void the argument; arbitrarily large artificial vacuum machines sit outside it, priced. Within known physics and ordinary environments, nothing survives the conjunction. The universe, on current evidence, does not ship rocks at lightspeed — not because no one has tried hard enough, but because every door in the shipping department is on fire, half a c too slow, or a thousand AU long.

References

1. Ackeret, J. (1946). "Zur Theorie der Raketen." *Helvetica Physica Acta* 19, 103–112. (Relativistic rocket equation.)
2. Atwater, H. A., et al. (2018). "Materials challenges for the Starshot lightsail." *Nature Materials* 17, 861–867.
3. Aurell, E., Mejía-Monasterio, C., & Muratore-Ginanneschi, P. (2011). "Optimal protocols and optimal transport in stochastic thermodynamics." *Physical Review Letters* 106, 250601.
4. Cohen, A. N., et al. (2022). *The Astrophysical Journal* 930, 74. arXiv:2201.02721. (Blistering thresholds)

- and leading-edge erosion for relativistic spacecraft.)
5. Drobny, J., Cohen, A. N., Curreli, D., Lubin, P., Pelizzo, M., & Umansky, M. (2021). "Damage to relativistic interstellar spacecraft by ISM impact gas accumulation." *The Astrophysical Journal* 908, 248. arXiv:2103.07517.
 6. Early, J. T., & London, R. A. (2000). "Dust grain damage to interstellar laser-pushed lightsail." *Journal of Spacecraft and Rockets* 37, 526–531; London, R. A., & Early, J. T. (2018). *JBIS* 71, 133.
 7. Forward, R. L. (1984). "Roundtrip interstellar travel using laser-pushed lightsails." *Journal of Spacecraft and Rockets* 21, 187–195.
 8. Guillochon, J., & Loeb, A. (2015). "The fastest unbound stars in the universe." *The Astrophysical Journal* 806, 124. arXiv:1411.5022.
 9. Hayakawa, S. (1972). "Cosmic ray dust grains." *Astrophysics and Space Science* 16, 238. (Relativistic-dust UHECR hypothesis.)
 10. Hills, J. G. (1988). "Hyper-velocity and tidal stars from binaries disrupted by a massive Galactic black hole." *Nature* 331, 687–689.
 11. Hoang, T., Lazarian, A., & Schlickeiser, R. (2015). "On origin and destruction of relativistic dust and its implication for ultrahigh energy cosmic rays." *The Astrophysical Journal* 806, 255. arXiv:1412.0578.
 12. Hoang, T., Lazarian, A., Burkhart, B., & Loeb, A. (2017). "The interaction of relativistic spacecrafts with the interstellar medium." *The Astrophysical Journal* 837, 5. arXiv:1608.05284.
 13. Ilic, O., Went, C. M., & Atwater, H. A. (2018). "Nanophotonic heterostructures for efficient propulsion and radiative cooling of relativistic light sails." *Nano Letters* 18, 5583–5589.
 14. Karlušić, M. (2017). arXiv:1701.04319. (Unrefereed comment arguing ref. 12's ion-track damage model overestimates gas damage; carried here as a qualification.)
 15. Koposov, S. E., et al. (2020). "Discovery of a nearby 1700 km/s star ejected from the Milky Way by Sgr A*." *Monthly Notices of the Royal Astronomical Society* 491, 2465–2480. (S5-HVS1.)
 16. Kulkarni, N., Lubin, P., & Zhang, Q. (2018). "Relativistic spacecraft propelled by directed energy." *The Astronomical Journal* 155, 155. arXiv:1710.10732.
 17. Lubin, P. (2016). "A roadmap to interstellar flight." *Journal of the British Interplanetary Society* 69, 40–72. arXiv:1604.01356.
 18. Marx, G. (1966). "Interstellar vehicle propelled by terrestrial laser beam." *Nature* 211, 22–23.
 19. Purcell, E. M. (1963). "Radioastronomy and communication through space." Brookhaven Lecture Series BNL-658.
 20. Redding, J. L. (1967). "Interstellar vehicle propelled by terrestrial laser beam." *Nature* 213, 588–589.
 21. Semyonov, O. G. (2014). "Relativistic rocket: Dream and reality." *Acta Astronautica* 99, 52–70.
 22. Semyonov, O. G. (2018). "Pros and cons of relativistic interstellar flight." *Acta Astronautica* 151, 736–742. arXiv:1807.08608.
 23. Spitzer, L. (1949). "The dynamics of the interstellar medium. III." *The Astrophysical Journal* 109, 337. (Radiation-pressure grain acceleration.)
 24. von Hoerner, S. (1962). "The general limits of space travel." *Science* 137, 18–23.
 25. Wald, R. M. (1974). "Energy limits on the Penrose process." *The Astrophysical Journal* 191, 231–233.
 26. Yurtsever, U., & Wilkinson, S. (2018). "Limits and signatures of relativistic spaceflight." *Acta Astronautica* 142, 37–44. arXiv:1503.05845.

Citations 2–5, 8, 11, 12, 14–17, 21, 22, 26 were verified against their primary sources (abstract-level verbatim checks, 3-voter adversarial review) during the preparation of this paper; the remainder are standard historical references.

Appendix A: numerical verification chains

```
A1. gamma(0.99) = 1/sqrt(1-0.99^2) = 1/sqrt(0.0199) = 7.0888
A2. KE/kg = (7.0888-1)*(2.998e8)^2 = 6.0888*8.988e16 = 5.472e17 J/kg
A3. Reflected Doppler (1-0.99)/(1+0.99) = 0.01/1.99 = 1/199
A4. Sail-frame sweep sqrt(1.99/0.01) = sqrt(199) = 14.1
A5. E_inc = int_0^0.99 (1+b)/(2b) c^2 dgamma = 5.89e17 J/kg (numerical, N=2e6 steps)
A6. Absorbed at alpha=1e-6: 5.89e11 J/kg
A7. Sail 1 g/m^2 -> 1000 m^2/kg; P_rad = 2*eps*sigma*T^4*1000
    at eps=1, T=1000K: 2*5.67e-8*1e12*1000 = 1.134e8 W/kg
A8. t_min = 5.89e11/1.134e8 = 5.19e3 s = 1.44 h
A9. ISM flux (ship frame) = gamma*n*beta*c = 7.0888*1e6*0.99*2.998e8
    = 2.104e15 protons/m^2/s at n=1 cm^-3
A10. E/proton = 6.0888*938.27 MeV = 5.713 GeV = 9.153e-10 J
A11. P_ISM = 2.104e15*9.153e-10 = 1.926e6 W/m^2
A12. T_eq(2-face) = (1.926e6/(2*5.6704e-8))^0.25 = 2,030 K
A13. Dust: grain m = (4/3)pi(1e-7)^3*2000 = 8.4e-18 kg;
    E = 6.0888*8.4e-18*8.988e16 = 4.6 J
A14. Shield: 5*86 g/cm^2 = 430 g/cm^2 = 4,300 kg/m^2; /1e-3 kg/m^2 = 4.3e6x
A15. ISCO: r = 6GM/c^2; local v_circ = c/2 (Schwarzschild, any M)
A16. Tidal stress, 1-m rock, 10 Msun ISCO (r=88.6 km):
    2GM/r^3 = 2*1.327e21/(8.86e4)^3 = 3.82e6 s^-2
    sigma_t = 3000*3.82e6*0.25/2 = 1.4e9 Pa (FAILS: strength governs)
A17. Same rock, 4.3e6 Msun ISCO (r=3.81e10 m): 2GM/r^3 = 2.06e-5 s^-2
    sigma_t = 3000*2.06e-5*0.25/2 ~ 8e-3 Pa (margin ~1e10: ISCO governs)
A18. gamma(0.5) = 1.1547; KE ratio (gamma(0.99)-1)/(gamma(0.5)-1) = 39.4
A19. Rocket, photon: m0/m1 = gamma(1+beta) = 7.0888*1.99 = 14.1
A20. Rocket, fusion w=0.1c: log10(MR) = log10(199)/(2*0.1) = 11.49 -> 3.1e11
A21. Waste heat, eta=0.8: 5.472e17*0.25 = 1.368e17 J/kg
    t at 200 W/kg = 6.84e14 s = 2.17e7 yr
A22. Scissors aperture: D = 2.44*lam*<b>c</b>*alpha*E_inc*rho_A/(2 eps sigma T^4 s)
    baseline (1e-6,1.0,1000K,30m,1e-3): D = 117 km
    swept (1e-4,0.05,1000K,30m,1e-3): D = 2.3e5 km
A23. Mass-driver: a_max = S/(rho L) = 1e8/3000 = 3.33e4 m/s^2 = 3,400 g
    l = (gamma-1)c^2/a_max = 5.472e17/3.33e4 = 1.64e13 m = 110 AU
    (at ultimate strength; x10 at working stress -> ~1,100 AU)
A24. 10 pc transit at 0.99c: t = 10*3.086e16/(0.99*2.998e8) = 1.04e9 s = 33 yr
    fluence = 2.104e15 * 1.04e9 = 2.19e24 p/m^2
A25. Rapidity: w = atanh(beta); w(0.5) = 0.549, w(0.99) = 2.647
    -> 4.82 ideal 0.5c boosts reach 0.99c (chains are NOT kinematically capped)
A26. Proton rigidity at gamma=7.089: p = sqrt((gamma*m)^2 - m^2) = 6,585 MeV/c
    R = 6.58 GV; gyroradius r = p/(qB): 22 m at 1 T, 2.2 m at 10 T
A27. Chirped-transmitter endpoint: 1000 nm/14.1 = 71 nm; E_ph = 1239.8/70.9 = 17.5 eV
    (above every optical-material bandgap; LiF transmission cutoff ~105 nm)
A28. Chemical rocket w = 1.5e-5 c: log10(MR) = log10(199)/(2*1.5e-5) = 76,628
    at beta = 0.1/0.5/0.9: 2,905 / 15,904 / 42,625
A29. Fusion mass-ratio referent: 3e11 kg at rho=2000 -> sphere d = 659 m
    Ceres = 9.4e20 kg = 3.1e9 x larger; Ceres-mass vehicle delivers ~3e9 kg
A30. Low-density heating at 0.99c: n=0.01 cm^-3 -> 1.9e4 W/m^2, T_eq = 642 K
    n=1e-5 cm^-3 (halo/IGM) -> 19 W/m^2, T_eq = 114 K
```

Appendix B: corrections adopted during adversarial review

Positions tested and failed, recorded so they are not resurrected.

Round 1 (pre-draft review): (1) "transferred energy = deposited heat" — false, §2.2; (2) "any absorption at 6 mc² = vaporization" — false at design wavelength, §2.2; (3) "thin films can't be low-absorption mirrors" — false as stated (the true constraint is the three-way trade, §6); (4) "no surface survives the CW intensity" — false at

cm scale (the true constraint is defect statistics at sail area, §6.4); (5, 6) "gravitational acceleration/turning stresses the payload" — false, equivalence principle, §3 Lemma 2; (7) "spaghettification kills slingshotting rocks" — false around SMBHs — and the earlier draft's own tidal number for a $10 M_{\odot}$ ISCO was wrong in the favorable direction (corrected, §4.2: stellar-mass ISCOs do shred meter rocks; SMBH ISCOs do not); (8) "the stellar velocity ceiling applies to rocks" — false, scaling inverts, §4.2; (9) "thermal mass dooms rocks in flybys" — false (transient skin depth is cm-scale); (10) "surveys would show fast rocks" — zero evidentiary weight either way, §11; (11) "photon rocket and sail fail identically" — false, they fail by distinct inequalities (§8.2 vs. §§6–7).

Round 2 (external referee pass on draft 1; this draft): (12) "Chains asymptote; they do not diverge" — **false as kinematics**: boosts compound in rapidity (4.8 ideal 0.5c boosts reach 0.99c, A25); replaced by the mechanism closure of §4.3 (exchange capture collapses above the first boost's exit speed). (13) The ISCO speed is not a universal gravitational cap — it is the per-encounter scale of the bound-exchange channel only; §4 restructured channel-by-channel. (14) "A Ceres of fuel per kilogram" — wrong by 9+ orders of magnitude (3×10^{11} kg is a ~660-m asteroid, A29); fixed in §8.1. (15) Chemical rocket column header (1.5×10^{-5} c) disagreed with its numbers (computed at 5×10^{-5} c); both now use real chemical exhaust (A28). (16) "The acceleration itself destroys the [ppm absorption]" — overstated: a chirped transmitter evades the sail-frame sweep; replaced by the conservation-of-burden statement of §6.1 (the $14\times$ band lands on sail or source; the source endpoint is 71-nm VUV, A27). (17) "Six times its own annihilation energy" — ambiguous by $2\times$; now "6.09 times its rest-mass energy." (18) The headline claim rescoped from unconditional acceleration no-go to acceleration-plus-transport through ordinary interstellar environments, with the [I]/[P]/[E] modality promoted into the claim statement itself (§1.1–1.2) — the referee's single most important point, accepted in full. (19) "Mean-density" wording replaced by explicit density/column parameterization, with the low-density escape priced (§5.1, §9.5). (20) Active shielding was unaddressed; now enumerated and priced (§5.4).